MIT Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)	**COURSE SYLLABUS**	
SCHOOL OF COMPUTER ENGINEERING	**W.E.F**	**AY: 2025 - 2026** (Rev. 2023(NEP))
SECOND YEAR BACHELOR OF TECHNOLOGY INFORMATION TECHNOLOGY	**COURSE NAME**	Database Management System
	COURSE CODE	2308216T
	COURSE CREDITS	3
RELEASE DATE : 01/07/2025	**REVISION NO.**	0.0

TEACHING SCHEME (HOURS/WEEK)		**EXAMINATION SCHEME AND MARKS**					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
3	NIL	30	20	50	NIL	NIL	100

PREREQUISITE KNOWLEDGE : Foundation of Computing

COURSE OBJECTIVES :

2308216T.CEO.1: To learn basic concepts of database management system,database transaction processing,concurrency control and Recovery.
2308216T.CEO.2: To learn key notions of query evaluations and optimization techniques.
2308216T.CEO.3: To develop database design using ER diagram.
2308216T.CEO.4: To develop database design with integrity constraints and Trigger.
2308216T.CEO.5: To understand the normalization techniques.
2308216T.CEO.6: To understand CRUD operations on SQL and NoSQL database

COURSE OUTCOMES :

After successful completion of the course, students will be able to,

2308216T.CO.1: Illustrate the fundamental concepts of database management system, transaction processing,concurrency control and recovery. [L2]
2308216T.CO.2: Apply notions of query evaluations and optimization techniques. [L3].
2308216T.CO.3: Construct the ER model to solve the given problem[L3].
2308216T.CO.4: Design a well–formed relational database design(Schema) [L3]
2308216T.CO.5: Apply Normalization techniques to relational Schema[L3].
2308216T.CO.6: Execute CRUD operations on relational and non-relational database.[L3]

COURSE ABSTRACT:
A database management system is a software which allows creation,definition,and manipulation of the database,allowing users to store,process and analyse data easily. The course allows the learners with the learning of complete end to end development of an application developed using the UI and back end technologies.The data has to be mapped to different data models like the ER diagram,relational data model and then this has to be stored back into the database system.For effectively analyzing the stored data some query language is used. Here, in this course, the ORACLE or MYSQL can be used as a database system for storing and managing the data.The database must be created in such away that the data must be retrieved in an efficient manner. For this, the concept of normalization is applied to the database table. Finally, the ACID properties of any database application have to be maintained form a intaining the consistency of the data. The main objective of this course is to apply the basic operation of the database management system on the data generated through some application. Finally,the students must develop an end to end application considering some user requirements by applying front-end and back-end technologies.

THEORY COURSE CONTENTS		
UNIT 1	**INTRODUCTION**	**08 HOURS**
Applications/Case Study: Banking system,Student Information system using traditional file processing system **Contents:** Introduction to Database system,Database System Vs file System,Data Abstraction, Instances and Schemas,Data Models- the ER Model,Relational Model,Other Models Mapping ER model to relational model. Case study ERD and Table design,System Structure,Database architectures:Centralized,Client Server,Parallel and Distributed Systems. **Self Study:** Database Users and Administrator **Further Readings:** Modeling concept for object oriented and object relational database.		
UNIT 2	**RELATIONAL MODEL**	**08 HOURS**
Applications/Case Study: Relational model for Banking system,University database **Contents:** Structure of Relational Databases ,Database Schema,Schema Diagrams , Integrity Constraints over Relations-Key Constraints,Foreign Key Constraints,General Constraints,Relational Query Languages–Relational algebra,Tuple relational calculus. **Self Study:**:Equivalence of relational calculus and relational algebra. **Further Readings:**Domain relational calculus.		
UNIT 3	**STRUCTURE QUERY LANGUAGE**	**08 HOURS**
Applications/Case Study: SQL queries for Banking system, shop management system,Tiwtter data analysis **Contents:** SQL Queries–Nested queries–Aggregate operators–Null values,Views,Index,PL/SQL block,exceptions,packages, looping,Concept of stored procedures,cursor,Triggers. **Self Study:** Transaction control language–commit,Rollback,savepoints. **Further Readings:**Recursive Queries		

UNIT 4	DATABASE DESIGN AND QUERY PROCESSING	07 HOURS

Applications/Case Study: Student Information system,Employee database system
Contents: Relational Database Design: Purpose of Normalization,Data Redundancy and Update Anomalies,Functional Dependencies. Decomposition–Armstrong's axiom. The process of Normalization: 1NF,2NF,3NF,BCNF.Query Processing:Overview,Measures of Query cost,Selection and Join operations,Evaluation of Expressions.
Self Study: Multi valued dependency,4NF
Further Readings:XML and web databases

UNIT 5	TRANSACTION MANAGEMENT	08 HOURS

Applications/Case Study: ATM system,Banking system
Contents: Basic concept of a Transaction, ACID Properties, Database Architecture, Concept of Schedule, Serial Schedule. Serializability: Conflict and View, Cascaded aborts Recoverable and Non-recoverable Schedules. Concurrency Control:Need Locking methods,Deadlocks,Time stamping Methods. Different crash recovery methods: Shadow-Paging, Log-based Recovery: Deferred and Immediate,CheckPoint.
Self Study: Buffer management and remote backup
Further Readings: ARIES Recovery

UNIT 6	NOSQL DATABASES	06 HOURS

Applications/Case Study: Aadhar UIDAI system
Contents: Introduction to NOSQL database,Types of NoSQL Databases,ACID vs BASE,SQL vs NoSQL,MongoDB:Data Types,Documents,Collections,Database,CRUD Operations,Aggregation
Self Study: Pipeline and Indexing.
Further Readings: Multimedia Databases..

TEXT BOOKS

1. Silberschatz A.,Korth H.,Sudarshan S.Database System Concepts. 6th Edition,McGrawHill Publishers,2006, ISBN978-0-07-352332-3.
2. Elmasri R., Navathe S., Fundamentals of Database Systems.4th Edition, Pearson, Education, 2003, ISBN8129702282D.
3. Pramod J Sadalage,Martin Fowler,NoSQL-Abrief guide to the Emerging World of Polyglot Persistence Distilled.Addison-Wesley, ISBN978-0-321-82662-6.
4. Raghu ramaKrishnan,Johannes Gehrke DatabaseManagement Systems3/e,TMH

REFERENCE BOOKS

1. Rab P.Coronel C.Database Systems Design, Implementation and Management. 5th Edition,
2. Connally T.,Begg C .Database Systems3rd Edition,Pearson Education,2002,ISBN81-7808-861 4M.
3. Thoms on CourseTechnology,2002, ISBN981-243-135-7. .Date C., An Introduction to Database Systems 7th Edition,Pearson Education, 2002,ISBN81-7808-23
4. H Garcia-Molina, JD Ullman and Widom, Database Systems: The Complete Book 2ndEd.,PrenticeHall

E-RESOURCES

1. https://nptel.ac.in/courses/106106220
2. https://www.w3schools.com/sql/